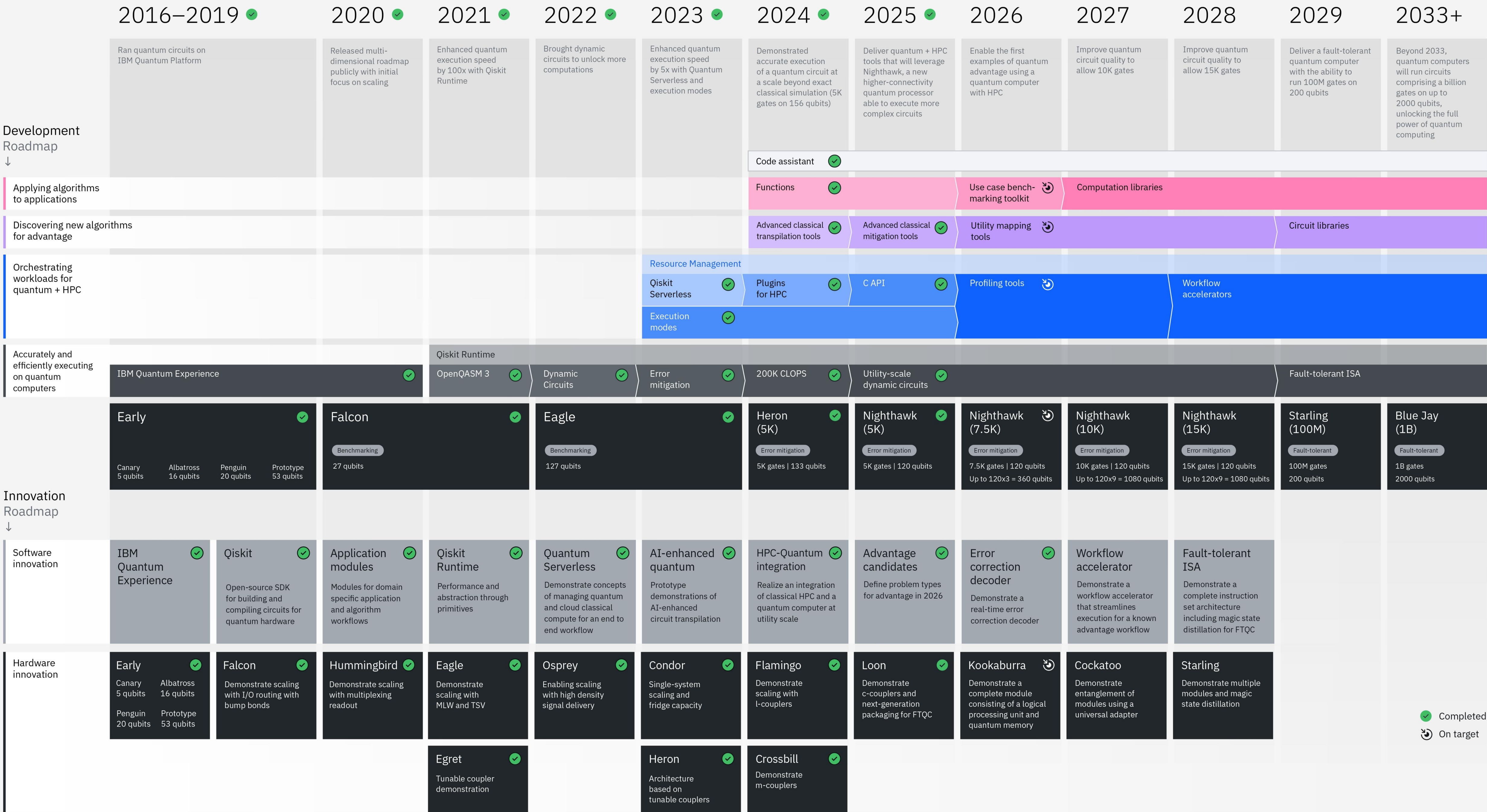




IBM Quantum Roadmap



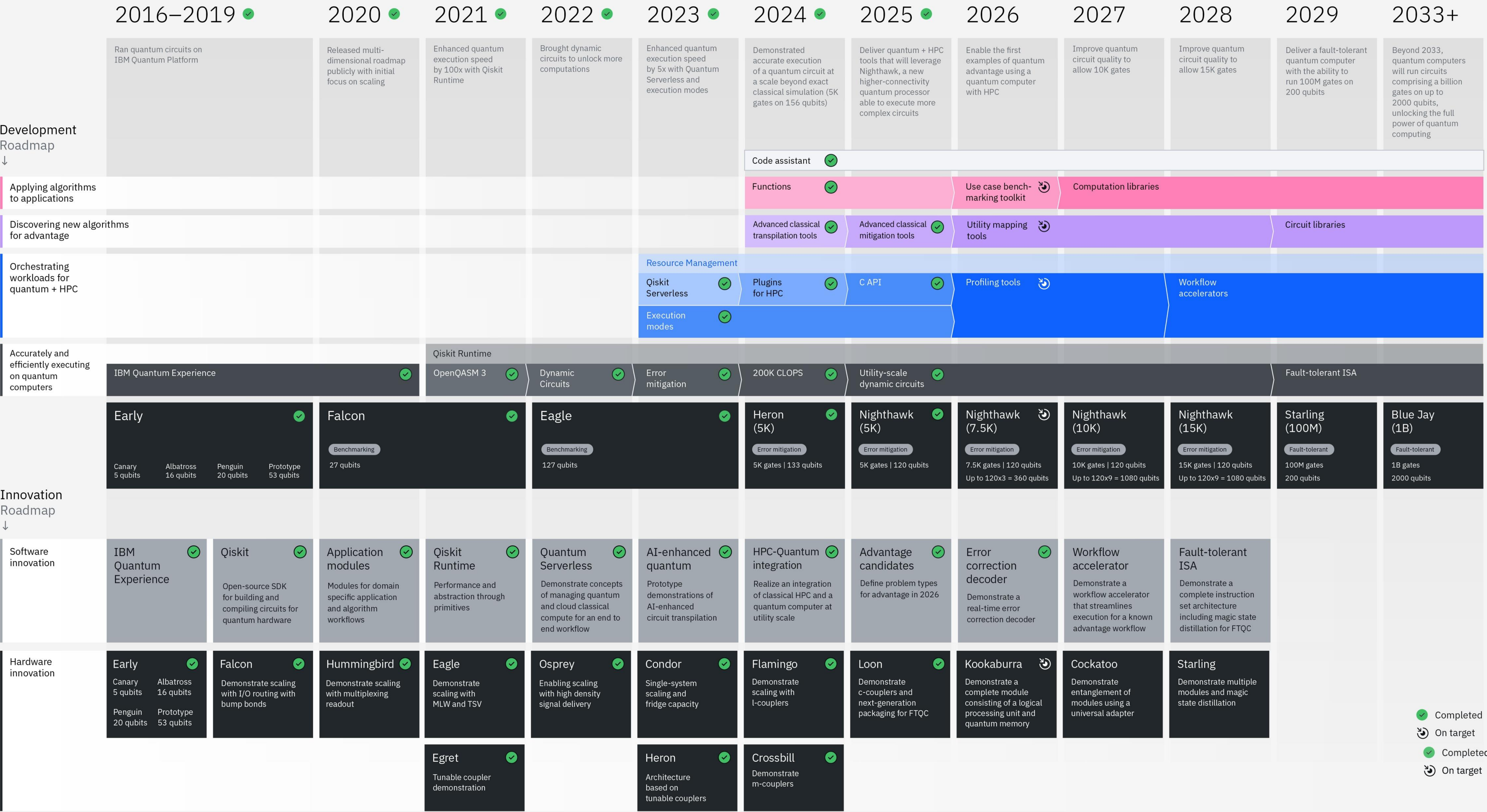
Completed

On target

Since 2020, we have tracked our plans to advance quantum computing with a public roadmap, checking off milestones as we achieved them.

In our 2023 roadmap, we committed to developing IBM Quantum Starling—the first large-scale, fault tolerant quantum computer—by 2029.

Our roadmap reflects two more years of progress toward that milestone, with the most recent updates made in January 2026.



Our roadmap is split into two parts:

Development Roadmap

Tracks the release of production hardware, software, and services delivered to users.

Innovation Roadmap

Charts the scientific breakthroughs and internal releases that underpin development milestones. Some items will remain internal proofs-of-concept, while others may evolve into future external releases.

Development Roadmap→

Development Roadmap:
Hardware

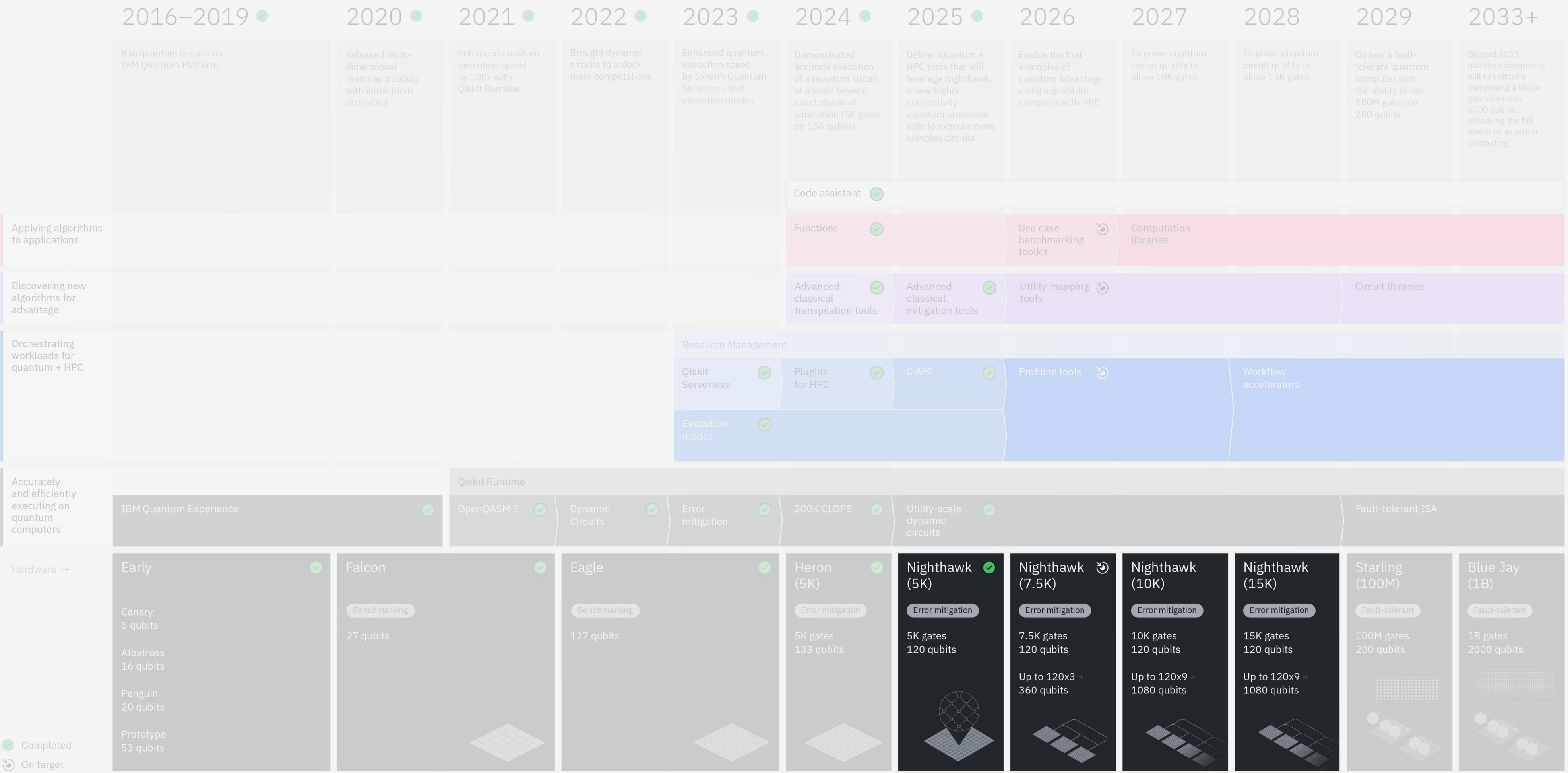
IBM Quantum Nighthawk is our platform for exploring and scaling quantum advantage ahead of large-scale fault-tolerant quantum computing.

Earlier quantum processors use a “heavy-hex” lattice, with each qubit connected to up to three neighbors. Nighthawk moves us to a square lattice, connecting each qubit to up to four neighbors.

This square lattice supports more efficient circuits, with fewer gates required for information routing. As a result, users can run more complex algorithms at the same gate scale.

Paired with techniques that reduce errors, Nighthawk is expected to run circuits with 5,000 gates by the end of 2025, 7,500 gates in 2026, 10,000 gates in 2027, and 15,000 gates in 2028.

As we improve and scale Nighthawk, we expect our team and partners to make major progress toward quantum advantages.



Completed
On target

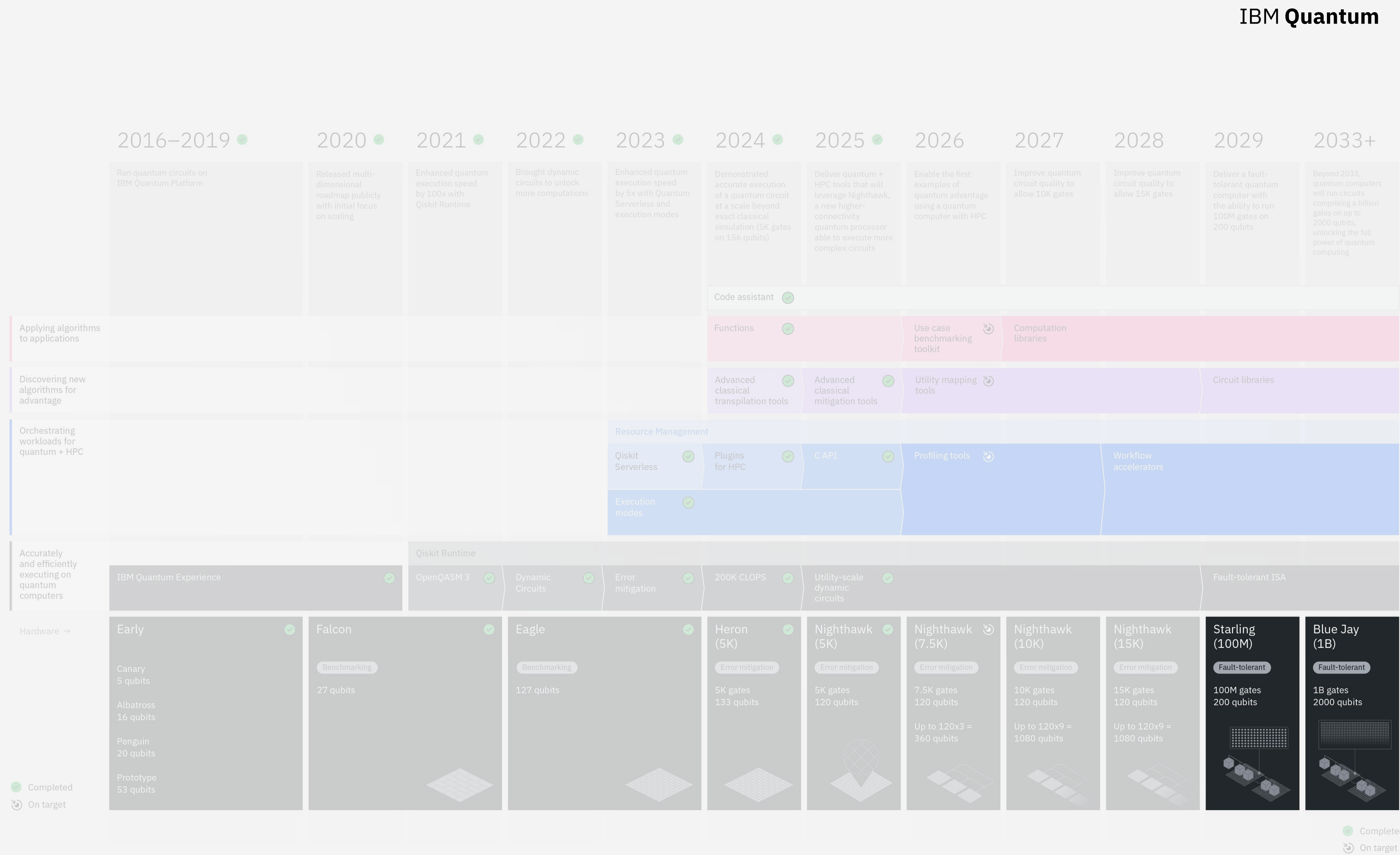
Completed
On target

Development Roadmap: Hardware

At the same time, we are driving toward **IBM Quantum Starling**—a system capable of running circuits with 100 million gates on 200 qubits—and aim to debut it by 2029.

In 2033, we will debut **IBM Quantum Blue Jay**, a system capable of running circuits with a billion gates on 2,000 qubits. These large-scale fault-tolerant quantum computers will unlock a new era of algorithmic complexity and application discovery.

Developers will not need to change how they write quantum programs in this era. They will simply notice that they can run longer workloads.



Development Roadmap:
Software, orchestration,
and execution

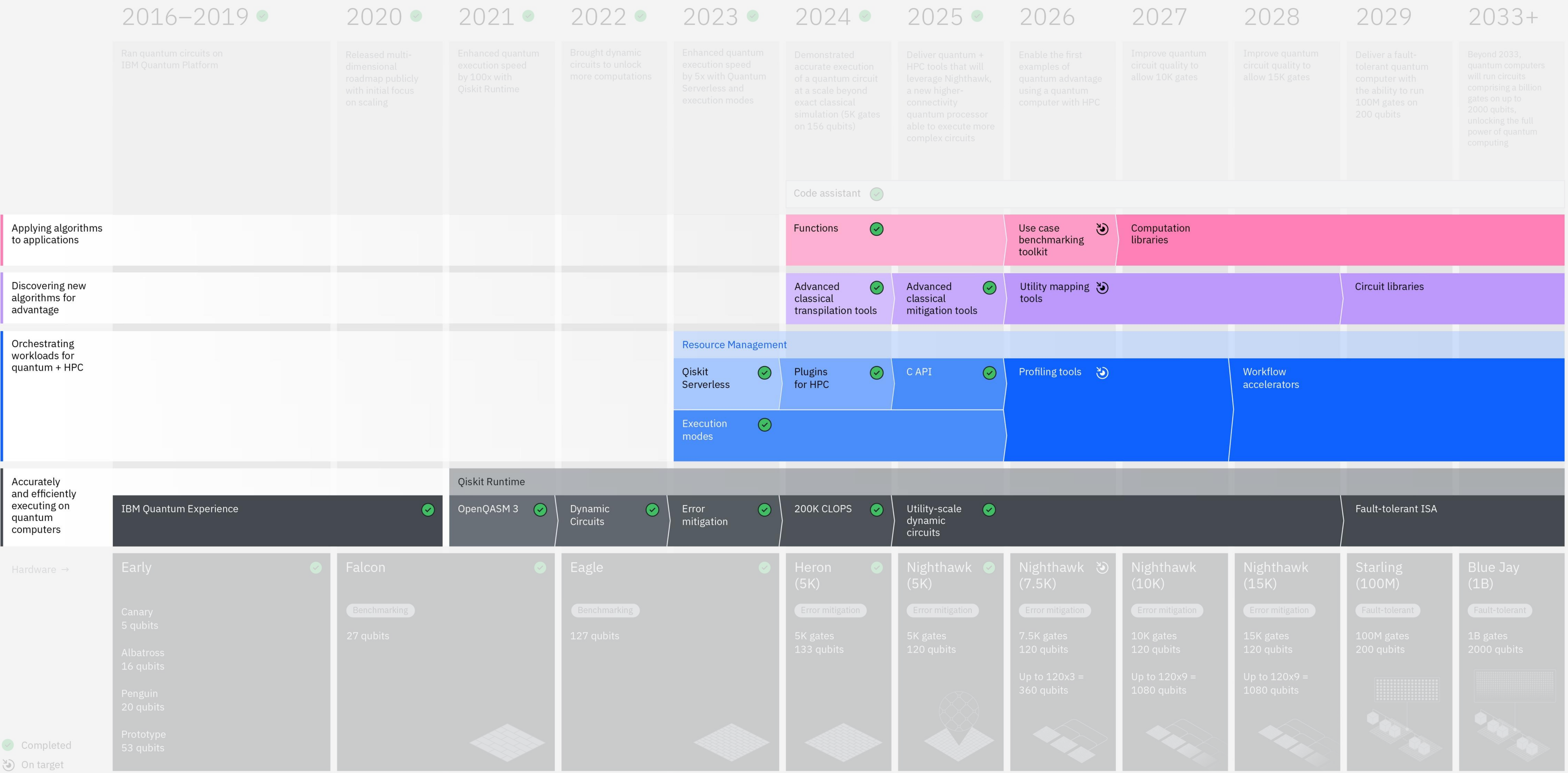
Useful quantum computing requires high-performance software. Our roadmap outlines four broad software objectives:

Apply algorithms to applications. These advances will let users adapt existing algorithms to domain-specific use cases.

Discover new algorithms for advantage. To move quickly to advantage, we must continue to improve and discover new algorithms. These advances provide a roadmap for tools that will enable new algorithm discovery.

Orchestrate workloads for quantum & high-performance computing (HPC). No algorithms are purely quantum, and we can more efficiently scale algorithms when we leverage quantum and classical resources together. So we are developing tools to better integrate and use quantum and classical resources.

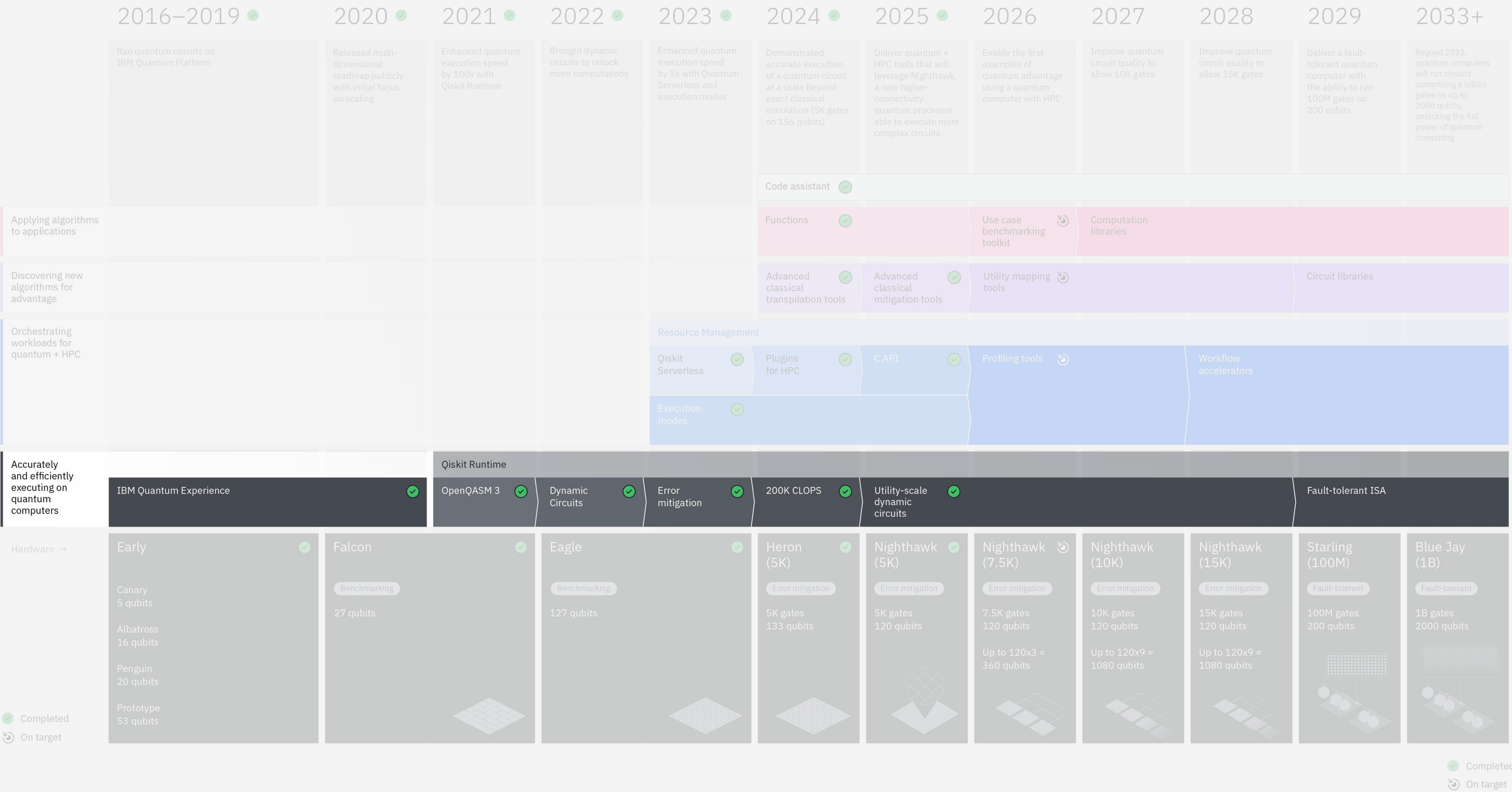
Execute accurately and efficiently. We will continue to deliver high-performance software that runs circuits accurately on our evolving quantum hardware.



Development Roadmap:
Execute accurately
and efficiently

In 2025, we introduced utility-scale dynamic circuits, extending the horizon of what can be accomplished on our existing quantum computers. This will allow us to run even larger and more complex circuits, which are required to realize useful quantum computing.

The work continues, even after a box is checked off on our roadmap. We will continue to deliver improvements to dynamic circuits, error mitigation, and speed in the coming years, further extending the capabilities available to our users.



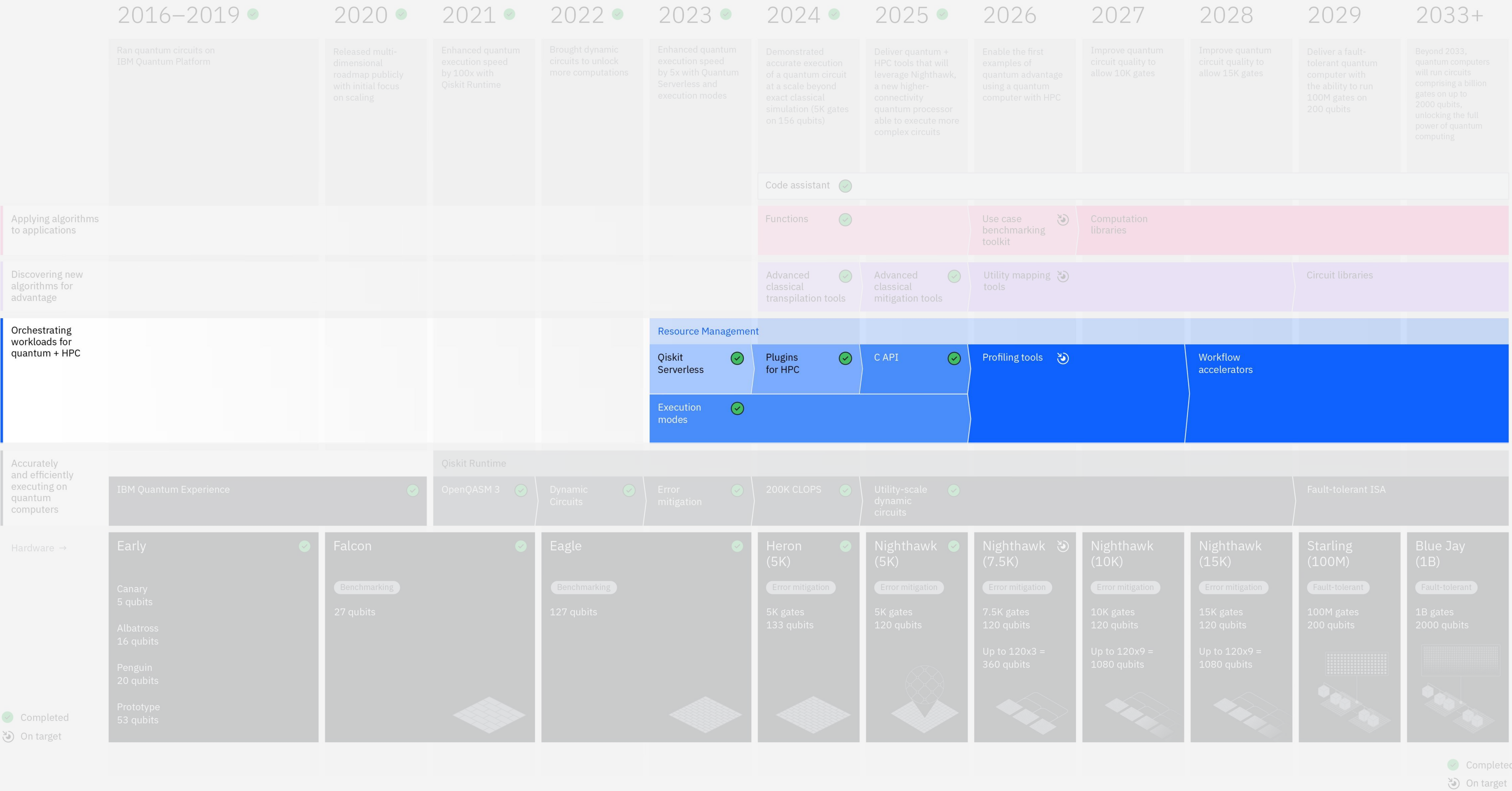
Development Roadmap:
Orchestrating workloads
for quantum & HPC

Running quantum workloads requires infrastructure that coordinates quantum resources with scalable classical resources.

In 2025, we released the C-API, enabling users to write Qiskit code in C, which is widely used for HPC. Soon, users will be able to write quantum and classical code in the same language and deploy it in an integrated system.

Moving into 2026 and beyond, we will introduce new profiling tools to help users monitor, verify, and debug workloads across quantum and classical resources.

In 2027 we will introduce workflow accelerators that deliver optimized quantum-classical execution pipelines for efficiently running similar tasks.



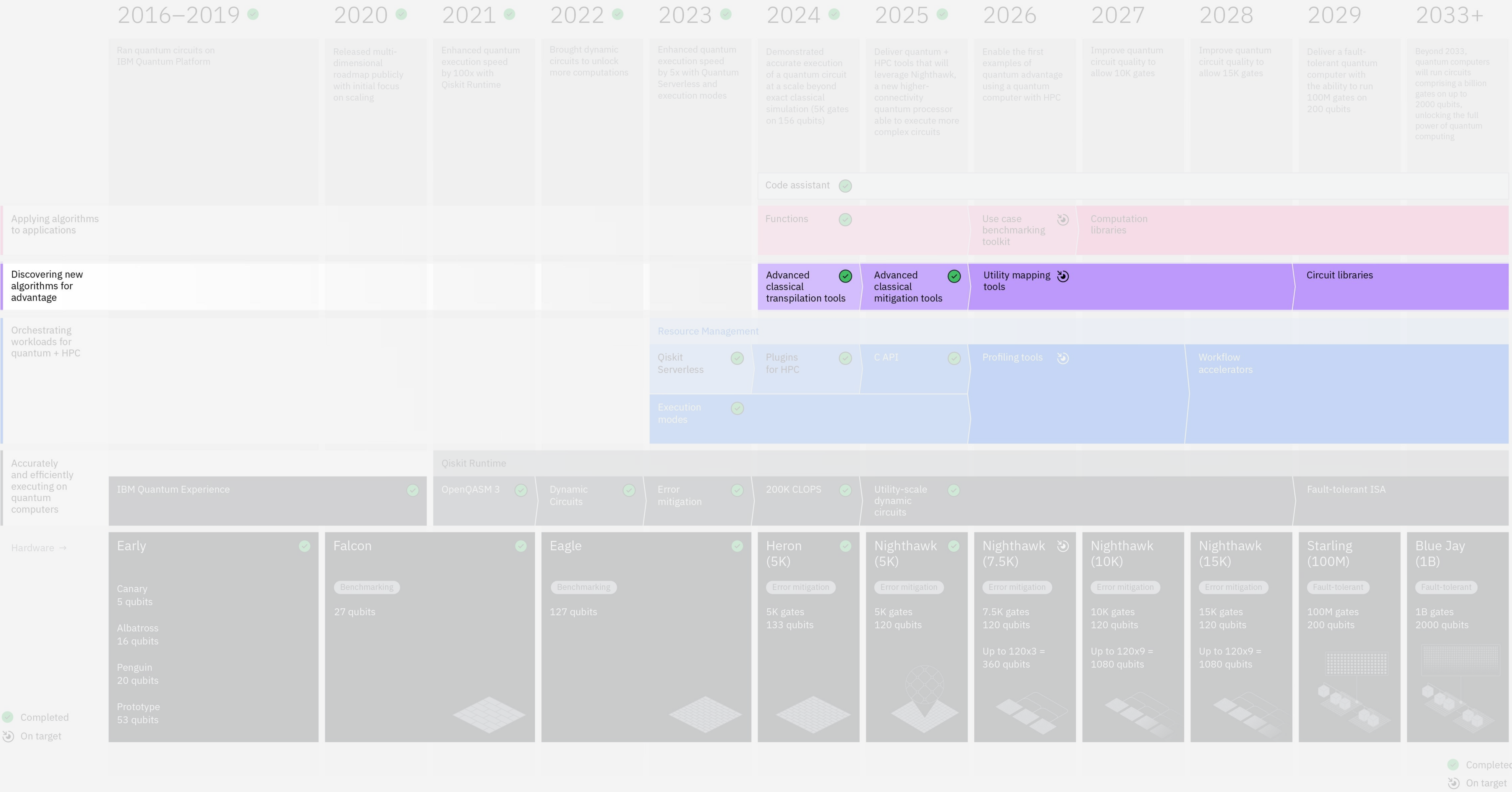
Development Roadmap:
Discovering new
algorithms for advantage

In 2024, we saw the emergence of powerful tools like AI transpilation passes and sample-based quantum diagonalization (SQD), which operate outside the runtime and use classical resources to suppress and mitigate errors.

In 2025, these tools matured further. We introduced error mitigation capabilities that leverage advanced classical subroutines to mitigate errors outside the runtime. These will help users identify circuit types and characteristics that scale prior to fault tolerance.

In 2026, we will introduce utility mapping tools. These support the exploration and design of new algorithms by mapping problems to circuits that also scale prior to fault tolerance.

By 2029, we expect to provide libraries that offer an expanded, generalized set of circuits ideal for execution on fault-tolerant processors.

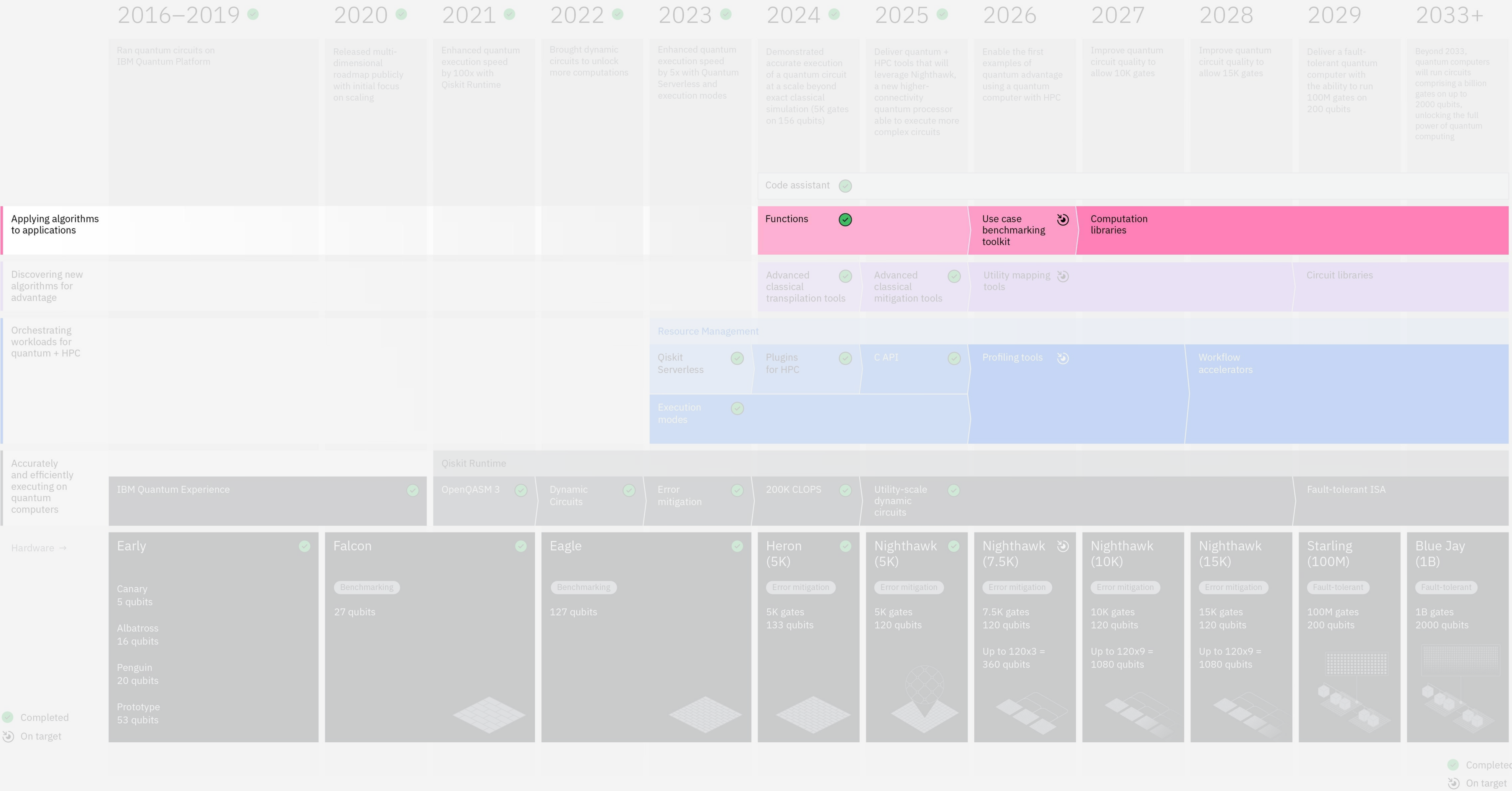


Development Roadmap:
Applying algorithms
to applications

In 2024, we successfully delivered the Qiskit Functions Service to our Premium Plan users. In 2025, we enhanced it with new functions.

Additionally in 2025, we established candidates for quantum advantage, as planned on our innovation roadmap. In 2026, we will work with partners to create a use case benchmarking tool, enabling others to explore which of their applications are ripe for near-term quantum value.

In 2027, we will introduce computation libraries. These provide abstractions—mathematical subroutines for applications—that integrate with popular existing computational libraries. This will support collaboration, integration, and help users more efficiently build and orchestrate workflows across classical and quantum compute resources.



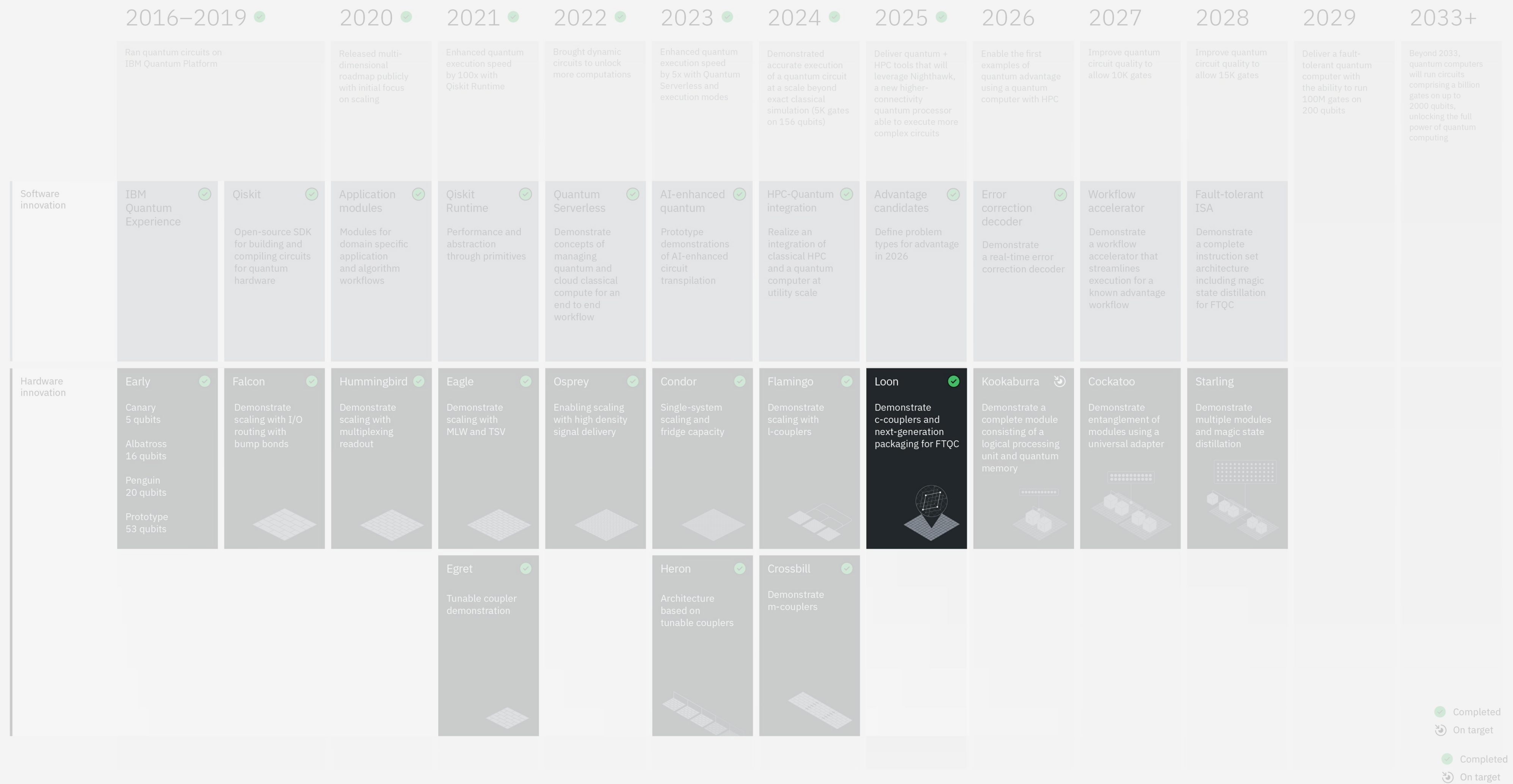
Innovation Roadmap→

Innovation Roadmap: Hardware

IBM Quantum Loon features a new chip architecture that leverages c- couplers to link qubits across the chip, beyond nearest neighbors. It enables up to six degrees of connectivity between qubits.

Capitalizing on this improved connectivity, IBM has developed a scalable error-correcting code that outperforms earlier methods. This advancement strengthens our confidence in achieving large-scale, fault-tolerant quantum computing by 2029.

IBM Quantum



Innovation Roadmap:
Software, algorithms,
and workflows

In 2025, we shortlisted candidate algorithms to demonstrate quantum advantage in 2026 and prototyped our error correction decoder ahead of schedule. This decoder will enable real-time error correction—a key capability for scalable, fault- tolerant quantum computing.

In 2026, we aim to demonstrate the first examples of quantum advantage using a quantum computer with HPC.

In 2027, we will prototype workflow acceleration strategies to streamline executions for known advantage workflows. This will help optimize execution across quantum and classical resources, saving developers valuable compute time.

In 2028, we will prototype a fault-tolerant instruction set architecture on our Starling proof-of concept, to be released the following year.

